

SNEHA KARNA

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SUMMARY

Undergraduate engineering student with a solid background in electronics, software development, and project coordination. Experienced in managing technical teams, executing large-scale events, and leading creative public relations and branding campaigns for premier student tech societies. Proven leader serving as Vice President and Creative Lead, skilled in cross-functional collaboration and dedicated to deploying impactful technology and automation systems.

EDUCATION

Tribhuvan University, Institute of Engineering, Purwanchal Campus

Feb 2024 - Present

Bachelor of Engineering in Electronics, Communication, and Information Engineering

Dharan, Nepal

- Successfully coordinated and managed historic student events including CS50 AI Nepal and X-Hack 3.0.

BKVM

Apr 2021 - Sep 2023

Secondary Level Education (10+2)

Biratnagar, Nepal

- Major courses in Physics, Chemistry, Mathematics, and Computer Science.

EXPERIENCE

EXCESS (Electronics & Communication Engineering Student Society)

March 2026 - Present

Vice President - 13th Committee

Dharan, Nepal

- Elected to executive leadership to steer strategic planning, financial oversight, and execution of comprehensive technical, cultural, and community engagement events.
- Coordinate closely with departmental heads to host premium workshops, seminars, and hackathons, ensuring an optimal competitive environment for 500+ student members.
- Strengthen community outreach and industrial connections by implementing robust communication channels and institutional networking campaigns.

Taranga: The Wave of Technology

Jan 2026 - Feb 2026

Creative Lead

Dharan, Nepal

- Spearhead visual identity, thematic branding layouts, and public-facing design frameworks for flagship technological projects.
- Manage multi-platform official media channels, crafting highly engaging technical content that boosts digital outreach, conversion, and user participation.
- Collaborate directly with hardware and software engineering teams to translate complex engineering systems into intuitive visual narratives.

CS50 AI Nepal

Oct 2024 - Jan 2025

Class Organizer & Peer Mentor

Dharan, Nepal

- Assisted in conducting formalized CS50 AI classes, simplifying core artificial intelligence, search algorithms, and machine learning principles through interactive sessions.
- Mentored academic peers in project-building frameworks, debugging paradigms, and code compilation mechanics.
- Coordinated the grand CS50 Fair, facilitating project demonstrations to an audience of over 200+ technology enthusiasts, academics, and industry representatives.

X-Hack 3.0 (EXCESS)

Nov 2024 - Jan 2025

Event Organizer & Logistics Lead

Dharan, Nepal

- Successfully executed one of the largest hardware and software hackathons in the university circuit, bringing together more than 150 active participants.
- Managed rigorous logistical workflows, team distributions, mentor schedules, and hardware asset provisions to maintain uninterrupted 48-hour development sprints.

PROJECTS

Driver Drowsiness & Safety Alert System

2026

Computer Vision Developer

Dharan, Nepal

- Built a real-time Computer Vision system using Python, OpenCV, and MediaPipe to monitor and mitigate driver fatigue.
- Implemented facial landmark detection algorithms to calculate eye aspect ratios (EAR) and mouth opening metrics for rapid drowsiness classification.
- Configured automated threshold validation routines to trigger safety alert mechanisms upon early fatigue detection.

Women Safety and Journey Monitoring System (Minor Project)

Ongoing

Hardware & Embedded Systems Developer

Dharan, Nepal

- Developing an IoT-driven real-time smart safety ecosystem embedded with GPS tracking and SOS trigger logic to safeguard transit environments.
- Integrating microcontroller processing units with wireless telemetry modules to seamlessly send geolocation logs and distress warnings to verified emergency contacts.
- Building a user-friendly mobile application interface using Flutter to dynamically plot paths and monitor live journey metrics.

Solar-Powered Smart Insect Trap (Delta 5.0)

Feb 2025

IoT Developer

Dharan, Nepal

- Engineered a sustainable agricultural IoT system that reduced pest infestations by 70% inside controlled ecological trial testbeds.
- Programmed advanced sensor automation scripts using C and Python for real-time sensor metric analysis and automated tracking parameters.
- Designed a low-power system architecture that optimized energy depletion, securing a 50% increase in overall battery cycle efficiency.

Basic Robotics and Microcontroller Hardware Setup

Feb 2024

Technical Specialist - Robotics Club IOEPC

Dharan, Nepal

- Engineered custom sensor-driven automation circuits with explicit logic parameters using microcontrollers, ultrasonic sensors, and motor driver infrastructure.

SKILLS

Programming Languages: C, C++, Python, Dart

Web & Mobile Technologies: HTML, CSS, Flutter (App Development)

Hardware & IoT: Embedded Systems, Arduino IDE, Raspberry Pi, Sensor Integration, Automation, Controls

Tools & Management: Git/GitHub, VS Code, Content Creation, Team Leadership, Social Media Branding, Public Relations

HOBBIES & PASSIONS

Content Creation: Translating advanced electronics, coding paradigms, and hardware engineering concepts into lucid, accessible technical blogs.

Event Management: Deeply passionate about formatting, hosting, and coordinating university-level technological hackathons and bootcamps.

Performing Arts: Channeling creative thinking and choreographic expression through freestyle and cultural dance.

LANGUAGES

Nepali (Native), English (Highly Proficient), Hindi (Highly Proficient)